

FROBENIUS CERTIFICATES FOR THE GALOIS GROUP OF $x(x-1)\cdots(x-p+1)+1$

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ABSTRACT. Let p be an odd prime and $f_p(x) = x(x-1)\cdots(x-p+1)+1$. An odd fixed-point-free Frobenius certifies $\text{Gal}(f_p/\mathbb{Q}) = S_p$, using a classification result for transitive groups of prime degree. At every prime $q < p$, the fixed-point-free condition is automatic, so a negative discriminant symbol suffices. We derive a reduced resultant formula and prove periodicity in p for each fixed q . The exact good classes for $q = 3, 5, 7$ certify a set of primes of Dirichlet density $7117/8064$; their joint distribution is not independent. The identification is verified for every odd prime $p < 10^7$, with explicit certificates for $p \geq 5$ and a separate cubic argument. A classical Jordan–Dedekind certificate covers $7 \leq p < 1500$ without classification. For $f_{p,c} = (x)_p + c$, $p \geq 5$ and $p \nmid c$, a window between the two smallest critical magnitudes gives exactly two non-real roots and hence S_p . The all-prime identification remains a conjecture; neither finite computations nor marginal densities alone settle it.

1. INTRODUCTION

Schur posed the irreducibility problem for products of distinct integral linear factors plus or minus one [13]; Westlund and Flügel settled it [22], with later extensions surveyed in [15]. The question studied here is the Galois group of the consecutive falling-factorial polynomial

$$f_p(x) = (x)_p + 1, \quad (x)_p = \prod_{j=0}^{p-1} (x-j),$$

for odd primes p . We distinguish the following conjecture from the certificates proved below:

Conjecture 1. *For every odd prime p , $\text{Gal}(f_p/\mathbb{Q}) = S_p$.*

The main contribution is an explicit method of certification for this family, rather than a proof of Conjecture 1. Theorem 3 converts an odd Frobenius with a root count different from one into S_p . Proposition 2 supplies zero roots at every $q \leq p$, and the reduced resultant formula (1) makes the discriminant symbol computable using polynomials of degree at most $q-1$. Periodicity then gives the exact coverage densities $1/2$, $31/40$, and $7117/8064$ for $q = 3$, $q \in \{3, 5\}$, and $q \in \{3, 5, 7\}$, respectively (Theorem 16). The first is obtained by hand; the additional densities use finite, reproducible computations.

The supplied witness list contains 664577 rows, one for every prime $5 \leq p < 10^7$. Its ordinary rows certify a negative symbol at a smaller odd prime; the row $p = 5$ uses a factorization at $q = 19$. Together with the separate

$p = 3$ argument, this proves the identification for all 664578 odd primes below the bound. Theorem 18 supplies an alternative classical certificate, recorded for all 236 primes $7 \leq p < 1500$. The root-count identity and critical-window theorem in §9 give a further application to $(x)_p + c$.

The classification-based input is isolated in Lemma 4; the Jordan certificate and critical-window theorem do not use it. The small Lean supplement proves only the affine fixed-point step. The computational certificates and their verification commands are specified in §10. The open questions in §11 require control of joint good sets, not merely their individual densities.

Related Galois-group methods include Newton polygons for generalized Laguerre polynomials [16] and criteria for trinomials [9, 10]; §8 explains the limitations of these routes for the prescribed specialization here. The prime-degree group list is classical [18, Thm. 2.1]; Lemma 4 spells out the parity consequence needed for the certificate. Jordan's theorem [21, Thm. 13.9] and the two-non-real-root criterion [1, Thm. 3.2] are likewise classical inputs. The contribution concerns their application to this family, its reduced discriminants, exact coverage, and critical-window counts.

2. THE POLYNOMIAL

Let p be an odd prime, $\varphi(x) = (x)_p = x(x-1) \cdots (x-p+1)$, and

$$f_p(x) = \varphi(x) + 1 \in \mathbb{Z}[x].$$

Reduction modulo p gives $f_p \equiv x^p - x + 1 \pmod{p}$, which is irreducible over $\mathbb{F}_p$ (Artin–Schreier). Hence f_p is irreducible over $\mathbb{Q}$, and $G = \text{Gal}(f_p/\mathbb{Q}) \leq S_p$ is transitive of prime degree.

The polynomial belongs to the family raised by Schur in 1908 [13]: is $(x - a_1) \cdots (x - a_n) \pm 1$ irreducible for distinct integers a_i ? Westlund and Flügel answered it the following year [22]: the -1 form is always irreducible, and the $+1$ form is irreducible unless it is, after a translation, $x(x-2)+1$ or $x(x-1)(x-2)(x-3)+1$. For the consecutive family this says that $(x)_n + 1$ is irreducible for every $n \neq 4$; see [15] for a modern account and the later generalisations. Irreducibility is therefore classical in every degree, and the Artin–Schreier argument above is quoted only to keep the prime-degree case self-contained. What is at issue here is the Galois group.

Proposition 2. *If $q \leq p$ is prime and $a \in \mathbb{F}_q$, then $f_p(a) = 1$. In particular f_p has no root in $\mathbb{F}_q$.*

Proof. The $p \geq q$ consecutive factors of $\varphi(a)$ cover every residue in $\mathbb{F}_q$, so $\varphi(a) = 0$ and $f_p(a) = 1$. $\square$

Thus Frob_q , for any prime $q \leq p$ not dividing $\text{disc } f_p$, is fixed-point-free (Dedekind). At $q = p$ the reduction $f_p \pmod{p}$ is irreducible, hence separable, so $p \nmid \text{disc } f_p$, and Frob_p is a p -cycle.

3. A GENERAL CERTIFICATE

Theorem 3. *Let $f \in \mathbb{Z}[x]$ be monic irreducible of degree p an odd prime, and let q be an odd prime with $q \nmid \text{disc } f$ such that*

- (1) *the number of roots of f in $\mathbb{F}_q$ is different from 1, and*
- (2) *$\left(\frac{\text{disc } f}{q}\right) = -1$.*

Then $\text{Gal}(f/\mathbb{Q}) = S_p$.

The second hypothesis is Stickelberger’s parity theorem: $\text{sgn}(\text{Frob}_q) = (\text{disc } f/q)$ [2, p. 186]; see also [3, Part I, Thms. 1 and 3].

Lemma 4. *Let p be an odd prime and $G \leq S_p$ transitive and nonsolvable. Then $G = S_p$ or $G \leq A_p$.*

Equivalently, the only transitive subgroups of S_p containing an odd permutation are S_p and certain subgroups of $\text{AGL}(1, p)$.

Proof of Theorem 3, given Lemma 4. Let $G = \text{Gal}(f/\mathbb{Q}) \leq S_p$, transitive, and $\sigma = \text{Frob}_q$. By Dedekind, σ has a fixed-point count different from one; by Stickelberger, σ is odd. If G were solvable, Galois’s theorem [18, §2] would embed it in $\text{AGL}(1, p) = \{x \mapsto ax + b\}$. The translations have order dividing the odd prime p , so they are even. Every odd affine map must therefore have $a \neq 1$, in which case it fixes exactly $b/(1 - a)$. This contradicts the fixed-point count of σ . Thus G is nonsolvable, so Lemma 4 gives $G = S_p$ or $G \leq A_p$; oddness of σ excludes the latter. $\square$

Remark 5 (The fixed-point condition). The simpler sufficient hypothesis that f has no root modulo q is often useful, and is automatic for f_p when $q \leq p$. The stronger formulation in Theorem 3 excludes every fixed-point count except one. For an S_p extension, as p tends to infinity the Chebotarev densities of odd derangements and of odd permutations whose fixed-point count differs from one tend to $1/(2e)$ and $(1 - e^{-1})/2$, respectively. Conditional on odd parity, these limits are e^{-1} and $1 - e^{-1}$. Parity remains a required hypothesis in both cases.

The affine argument is formalised in Lean 4 against `mathlib` in the supplied supplement. The two statements are

```
existsUnique_fixed_of_sign_eq_neg_one
sign_eq_one_of_not_existsUnique_fixed.
```

The source has no proof placeholders; its scope is this affine lemma, not the entire Galois-group certificate. The pinned toolchain and instructions for inspecting the axioms are in `formal/README.md`.

Remark 6 (Prime degree is essential). For example, the cyclic subgroup generated by (1234) is transitive and its generator is an odd derangement, so the conclusion of Theorem 3 fails at the first composite degree. The proof uses primality twice — Galois’s theorem for the solvable case, and Burnside’s for the rest — and both fail for composite n . By Westlund–Flügel the polynomials $(x)_n + 1$ are irreducible for every $n \neq 4$, so irreducibility is no longer the obstacle to a composite-degree statement; transitivity is not enough, and primitivity would have to be established separately.

Remark 7 (Sharpness). Neither hypothesis on σ can be dropped. A primitive root in $\text{AGL}(1, p)$ gives type $(1, p - 1)$, $\text{sign } (-1)^{p-2} = -1$; translations are fixed-point-free and even. Likewise A_p contains p -cycles. Only the conjunction “odd and fixed-point-free” is impossible outside S_p .

We first isolate the group-theoretic fact needed in the $\text{PSL}_2(11)$ case.

Lemma 8. *$\text{PGL}_2(11)$ has no subgroup of index 11.*

Proof. Suppose $H \leq \mathrm{PGL}_2(11)$ with $|H| = 120$. Since $120 \nmid 660$, one has $H \not\leq \mathrm{PSL}_2(11)$. The image of H in $\mathrm{PGL}_2(11)/\mathrm{PSL}_2(11) \cong C_2$ is therefore nontrivial, so $H/(H \cap \mathrm{PSL}_2(11)) \hookrightarrow C_2$ has order 2 and $H_0 = H \cap \mathrm{PSL}_2(11)$ has order 60. By Dickson's list [6, Hauptsatz II.8.27], the only subgroups of order 60 in $\mathrm{PSL}_2(11)$ are isomorphic to A_5 . Then $H \leq N := N_{\mathrm{PGL}_2(11)}(H_0)$ and $N \cap \mathrm{PSL} = N_{\mathrm{PSL}}(A_5) = A_5$ (A_5 maximal and non-normal), so $|N| \in \{60, 120\}$. If $|N| = 120$, the conjugation map $N \rightarrow \mathrm{Aut}(A_5)$ has kernel $C_N(A_5)$ meeting A_5 trivially, hence either

- $N = A_5 \times C_2$: then $A_5 \leq C_{\mathrm{PGL}_2(11)}(z)$ for an involution z . Every involution in $\mathrm{PGL}_2(11)$ fixes 0 or 2 points of $\mathbb{P}^1(\mathbb{F}_{11})$, so its centralizer lies in a torus normalizer of order $2(q \mp 1) \in \{20, 24\} < 60$.
- $N \cong S_5$: then N contains F_{20} , in which an element induces an automorphism of order 4 of C_5 . But a Sylow C_5 of $\mathrm{PGL}_2(11)$ lies in the split torus and fixes exactly two points, so $N(C_5)$ permutes that pair with kernel the diagonal torus C_{10} : order at most 20, image at most C_2 in $\mathrm{Aut}(C_5) \cong C_4$.

Thus $|N| = 60 < 120 = |H|$, a contradiction. $\square$

Proof of Lemma 4. By Burnside [7, §251], a transitive group of prime degree is solvable (hence affine) or 2-transitive. So G is 2-transitive, hence primitive. The socle of a 2-transitive group is regular elementary abelian or nonabelian simple; the abelian case is affine, excluded. Thus G is almost simple: $S \leq G \leq \mathrm{Aut}(S)$ with S nonabelian simple, transitive, and $[S : S_\omega] = p$. By Guralnick [5] (cf. Fengler [4, Cor. 2.39]), (S, S_ω) is one of:

- (1) $S = A_p$, $S_\omega = A_{p-1}$;
- (2) $S = \mathrm{PSL}_d(q_0)$, $p = (q_0^d - 1)/(q_0 - 1)$, S_ω a point or hyperplane stabilizer (d prime);
- (3) $S = \mathrm{PSL}_2(11)$, $S_\omega = A_5$, $p = 11$;
- (4) $S = M_{11}$ ($p = 11$) or $S = M_{23}$ ($p = 23$).

(The remaining case of Guralnick's theorem, $S = \mathrm{PSU}_4(2)$ with a subgroup of index 27 [4, Thm. 2.38], does not occur: $27 = 3^3$ is a prime power but not a prime.)

The restriction $\mathrm{sgn}|_S$ has kernel of index at most 2; a nonabelian simple group has no subgroup of index 2. So $S \leq A_p$ and sgn factors through $G/S \hookrightarrow \mathrm{Out}(S)$.

Case A_p . Then $G \in \{A_p, S_p\}$.

Case M_{11}, M_{23} . $\mathrm{Out}(M_{11}) = \mathrm{Out}(M_{23}) = 1$ [12], so $G = S \leq A_p$.

Case $\mathrm{PSL}_2(11)$. Here $\mathrm{Aut}(\mathrm{PSL}_2(11)) = \mathrm{PGL}_2(11)$, so $S < G$ would force $G = \mathrm{PGL}_2(11)$, whose transitive action on the 11 points would require a subgroup of index 11, contradicting Lemma 8. Hence $G = \mathrm{PSL}_2(11) \leq A_{11}$.

Case $\mathrm{PSL}_d(q_0)$, $q_0 = r^e$. Let $\Omega = \mathbb{P}^{d-1}(\mathbb{F}_{q_0})$ and take S_ω to be a point stabilizer (composing with the graph automorphism if needed). Distinct points of a 2-transitive group have distinct stabilizers, so if $c \in C_{\mathrm{Sym}(\Omega)}(S)$ then $S_{c(\omega)} = cS_\omega c^{-1} = S_\omega$, hence $c = 1$; the trivial centralizer is used in (c) below. Separately, since $S \trianglelefteq G$, we have $G \leq N_{\mathrm{Sym}(\Omega)}(S)$, and every $g \in G$ satisfies $gS_\omega g^{-1} = S_{g\omega}$, so g preserves the S -class of the point stabilizer.

(a) *No diagonal part.* As d is prime, $d \mid q_0 - 1$ would give $p \equiv d \equiv 0 \pmod{d}$, hence $p = d < 2^d - 1 \leq p$, absurd. Thus $\mathrm{PGL}_d(q_0) = \mathrm{PSL}_d(q_0)$.

(b) *No graph part, $d \geq 3$.* The graph automorphism swaps the point- and hyperplane-stabilizer classes [12, §3.3], while field automorphisms preserve the class of a point stabilizer. Class preservation therefore forces $G \leq \text{P}\Gamma\text{L}_d(q_0)$ and $G/S \leq \langle \varphi_r \rangle$.

(c) *The field part is even.* Any $g \in G$ inducing φ_r^k equals $s \cdot \varphi_r^k$ in $\text{Sym}(\Omega)$ with $s \in S \leq A_p$, so $\text{sgn}(g) = \text{sgn}(\varphi_r)^k$. If e is odd, φ_r has odd order, all cycles have odd length, and φ_r is even. This covers all $d \geq 3$: if e were even, $s_0 = r^{e/2}$ would give

$$p = \frac{s_0^d - 1}{s_0 - 1} \cdot \frac{s_0^d + 1}{s_0 + 1}$$

with both factors integers > 1 (d odd). For $d = 2$, $p = q_0 + 1$ prime forces $q_0 = 2^e$ with $e = 2^j$ (Fermat). On $\mathbb{P}^1(\mathbb{F}_{2^e})$, φ_2 fixes $\{0, 1, \infty\}$ and every nontrivial orbit has 2-power length ≥ 2 , hence is a cycle of even length and an odd permutation, so $\text{sgn}(\varphi_2) = (-1)^t$ with t the number of nontrivial orbits. The number of orbits of length 2^i is the number of monic irreducibles of degree 2^i over $\mathbb{F}_2$,

$$n_i = \frac{2^{2^i} - 2^{2^{i-1}}}{2^i}, \quad v_2(n_i) = 2^{i-1} - i.$$

Thus $n_1 = 1$ and $n_2 = 3$ are odd, while n_i is even for $i \geq 3$. For $j \geq 2$, $t \equiv n_1 + n_2 \equiv 0 \pmod{2}$ and φ_2 is even. For $j = 1$, $\text{P}\Gamma\text{L}_2(4) \cong S_5 = S_p$.

In all cases $G = S_p$ or $G \leq A_p$. $\square$

Remark 9 (Which p need the classification at all). Case (2) of the proof occurs only when $p = (q_0^d - 1)/(q_0 - 1)$ for some prime power q_0 and prime d — a *projective prime* in the terminology of Jones and Zvonkin [18], who observe that the classification of transitive groups of prime degree is complete except for the open question of which integers of this shape are prime, and give evidence that infinitely many are. If p is not a projective prime and $p \notin \{11, 23\}$, the only transitive subgroups of S_p are the subgroups of $\text{AGL}(1, p)$, A_p and S_p . For such p , the exceptional cases in the classification-based proof disappear, and Lemma 8 is unnecessary. The assertion that the displayed list exhausts the possibilities still uses classification; Burnside’s theorem alone does not identify all nonsolvable groups of prime degree. The projective primes below 100 are 3, 5, 7, 13, 17, 31 and 73 (those with $d = 2$ are exactly the Fermat primes, as in the proof above); together with 11 and 23 they are the only $p < 100$ for which the classification is invoked.

CFSG enters only through Guralnick’s theorem. The facts $\text{Out}(M_{11}) = \text{Out}(M_{23}) = 1$ and non-conjugacy of the two parabolic classes for $d \geq 3$ are in [12]; maximality of A_5 in $\text{PSL}_2(11)$ is in Dickson’s list [6, Hauptsatz II.8.27]. The involution-centralizer bound and the absence of F_{20} in $\text{PGL}_2(11)$, and the signs of φ_2 on $\mathbb{P}^1(\mathbb{F}_{2^e})$ for $e = 2, 4, 8, 16$, are finite checks. (If an order-4 element conjugated an element of order 5 to its cube, its inverse would conjugate it to its square, so testing $bab^{-1} = a^2$ over all order-4 elements covers both generators of $\text{Aut}(C_5)$.)

4. THE CERTIFICATE FOR f_p

Corollary 10. *Let p be an odd prime. If some odd prime $q < p$ satisfies $(\frac{\text{disc } f_p}{q}) = -1$, then $\text{Gal}(f_p/\mathbb{Q}) = S_p$.*

Proof. Hypothesis (1) of Theorem 3 holds for every prime $q \leq p$ by Proposition 2, so Theorem 3 applies. $\square$

The prime $q = p$ can never serve: Frob_p is an even p -cycle, equivalently $\text{disc } f_p \equiv (-1)^{(p+1)/2} \pmod{p}$ is always a residue. Evaluating the symbol at a candidate $q < p$ is a single resultant over $\mathbb{F}_q$. For such q ,

$$f_p \equiv (x^q - x)^{\lfloor p/q \rfloor} \prod_{k < p \bmod q} (x - k) + 1 \pmod{q},$$

and $\text{disc } f_p \bmod q$ is the discriminant of this reduction; the symbol is its quadratic character. A search over odd primes $q < p$ finds a witness $q \leq 73$ for every prime $7 \leq p < 10^7$ (largest witness $q = 73$ at $p = 9683099$; $q = 3$ succeeds for 50.003% of the 664577 primes, matching the density $\frac{1}{2}$ of the classes in Proposition 14 to five significant figures). For $p = 5$ the only smaller odd-prime candidate is $q = 3$, and $(38569/3) = +1$, so the certificate has no witness; S_5 is settled instead by the type-(2, 3) Frobenius at $q = 19$. For $p = 3$, $\text{disc } f_3 = -23$ is not a square, so $G = S_3$.¹

5. PERIODICITY OF THE DISCRIMINANT

The same identity that makes Frob_q fixed-point-free computes the Stickelberger symbol at a fixed odd prime $q < p$. Write $p = mq + r$ with $0 < r < q$, and set $g = x^q - x$, $B_r = (x)_r$. Then $\varphi \equiv g^m B_r \pmod{q}$. Since $r < q$, the factor B_r divides g ; write $g = B_r C_r$ with $C_r = \prod_{a=r}^{q-1} (x - a)$, and set

$$u_r := C_r B'_r,$$

a polynomial of degree $q - 1$ depending only on r . (In particular $u_1 = x^{q-1} - 1$.) Differentiating and using $g' \equiv -1$ gives

$$\psi := g B'_r - m B_r = B_r(u_r - m).$$

The roots of B_r lie in $\mathbb{F}_q$, so by Proposition 2 they contribute 1 to the resultant. Multiplicativity together with $\text{Res}(g, f_p) = \prod_{a \in \mathbb{F}_q} f_p(a) = 1$ yields

$$(1) \quad \text{disc } f_p \equiv (-1)^{(p-1)/2} r^p \prod_{u_r(\beta)=m} (g(\beta)^m B_r(\beta) + 1)^{\mu_\beta} \pmod{q},$$

the product over the fibre $u_r^{-1}(m)$ in an algebraic closure, with multiplicity. (The leading coefficient of u_r is r , since B_r is monic of degree $r < q$.) The unbounded part of the discriminant has disappeared. What remains is periodic, and depends on p only through $r, m \bmod q$, and the values $g(\beta)^m$ at finitely many β of degree at most $q - 1$.

Lemma 11 (Periodicity). *Let q be an odd prime and*

$$\Pi(q) = \text{lcm}\left(4, q^2, \text{lcm}_{1 \leq d \leq q-1} (q^d - 1)\right).$$

¹The complete main witness list for $5 \leq p < 10^7$, the $p \leq 97$ discriminant data, the $q = 3$ and $q = 5$ class tables of §5, the measured ε_q for odd $q \leq 199$, and the verification programs are included with this note (`ancillary/` for data, `tools/` for the checkers, one per computational claim). Each ordinary row (p, q) is independently verifiable by one resultant and one Legendre symbol over $\mathbb{F}_q$; the special $p = 5$ row is checked by its factorization.

For odd primes $p, p' > q$, if $p \equiv p' \pmod{\Pi(q)}$ then $\text{disc } f_p \equiv \text{disc } f_{p'} \pmod{q}$, and in particular $(\text{disc } f_p/q) = (\text{disc } f_{p'}/q)$.

Proof. The right-hand side of (1) is defined for any odd integer $n = mq + r > q$ not divisible by q , via $f_n = (x)_n + 1$. The sign $(-1)^{(n-1)/2}$ has period 4. The fibre polynomial $u_r - m$ is determined by r and $m \pmod{q}$, hence by $n \pmod{q^2}$; this is the only source of the factor q^2 in $\Pi(q)$. The power r^n depends on $n \pmod{q-1}$, and $q-1$ divides the $d=1$ term of the inner lcm.

Each factor $g(\beta)^m B_r(\beta) + 1$ is an evaluation in a field. If $\beta \in \mathbb{F}_q$ then $g(\beta) = 0$, so the factor is 1. If $\beta \notin \mathbb{F}_q$ then $g(\beta) \neq 0$, because every root of g lies in $\mathbb{F}_q$, and $g(\beta)^m$ has period $\text{ord}(g(\beta))$ dividing $q^d - 1$ where $d = [\mathbb{F}_q(\beta) : \mathbb{F}_q] \leq q-1$.

If $n \equiv n' \pmod{\Pi(q)}$, these data agree. (For the multiplicative orders: $\gcd(q, q^d - 1) = 1$, so $n \equiv n'$ modulo q^2 and modulo $\text{lcm}_d(q^d - 1)$ implies $m \equiv m'$ modulo that lcm.) Restricting to primes yields the lemma. $\square$

Remark 12. The quadratic fibres are more rigid. Always $g(\beta) = \beta^q - \beta$ lies in $\ker \text{Tr}_{\mathbb{F}_{q^d}/\mathbb{F}_q}$. For $d=2$ the trace-zero hyperplane coincides with a multiplicative coset: $\text{Tr}(x) = x + x^q = 0$ if and only if $x^{q-1} = -1$ for $x \neq 0$. Thus if $\beta \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$ and $\gamma = g(\beta)$, then $\gamma^q = -\gamma$, so $\gamma^{q-1} = -1$. In particular $\text{ord}(\gamma)$ divides $2(q-1)$ and does not divide $q-1$. (For $d \geq 3$ the trace-zero set is a plain additive hyperplane of size q^{d-1} , with no multiplicative structure, and nothing of this kind carries over.) The factor $q^2 - 1$ in $\Pi(q)$ may therefore be replaced by $2(q-1)$.

Remark 13 (Refined bounds and certified periods). Keep the general bound $\Pi(q)$ distinct from the least even period $P(q)$ of the symbol on odd integers $n > q$ coprime to q ; shifts preserve this domain. Let

$$E_q = \text{lcm}_{\beta: g(\beta) \neq 0} \text{ord}(g(\beta)), \quad M(q) = \text{lcm}(4, q^2, q-1, E_q),$$

where the lcm is over roots in all fibres. The proof of Lemma 11 shows that $M(q)$ is a period bound. The orders calculated in finite fields give

$$\begin{aligned} E_3 &= 4, & E_5 &= 48, & E_7 &= 5472, \\ E_{11} &= 62716735200, & E_{13} &= 50777730551520. \end{aligned}$$

The general bounds are

$$\Pi(3) = 72, \quad \Pi(5) = 483600, \quad \Pi(7) = 807353517600.$$

Using the specific orders instead gives $M(3) = 36$, $M(5) = 1200$, and $M(7) = 268128$. The certified symbol periods are

$$P(3) = 36, \quad P(5) = 600, \quad P(7) = 134064.$$

For $q=7$, the supplied finite certificate evaluates every odd domain class modulo the proven upper period 268128, verifies invariance under translation by 134064, and records a counterexample for each proper domain-preserving even divisor of 134064. Thus the positive period assertion is not based on a finite initial window. The program `tools/verify_periods.py` reproduces this certificate. Higher measured orders make direct class enumeration substantially more expensive; no impossibility of obtaining further exact densities by other methods is asserted.

6. EXACT CONGRUENCE COVERAGE

The bound is not always sharp: $\Pi(3) = 72$ while the actual period is 36. For $q = 3$ the fibres of $u_1 = x^2 - 1$ and $u_2 = 2(x + 1)^2$ can be read off by hand, and among odd p the value $\text{Res}(\psi, f_p)$ is constant on each class modulo 9.

Proposition 14. *Let $p > 3$ be prime. Then $(\text{disc } f_p/3) = -1$ if and only if*

$$p \equiv 7, 13, 17, 19, 23, 29 \pmod{36}.$$

Proof. Write $p = 3m + r$ with $r \in \{1, 2\}$ and $g = x^3 - x$. These are the fibres of $u_1 = x^2 - 1$ and $u_2 = 2(x + 1)^2$. The six values of $\psi = B_r(u_r - m)$ as $r \in \{1, 2\}$ and $m \pmod{3}$ are as follows. In each case $\text{Res}(\psi, f_p)$ is a constant $R \in \{1, 2\}$ on odd p , recorded with the corresponding class $p \pmod{9}$.

$r = 1$. Here $B_1 = x$, so $\psi = g - mx = x(x^2 - (m + 1))$.

- $m \equiv 0 \pmod{3}$: $\psi = g$. All roots lie in $\mathbb{F}_3$, so $f_p(\beta) = 1$ at every root and $R = 1$. Then $p \equiv 1 \pmod{9}$.
- $m \equiv 2 \pmod{3}$: $\psi = x^3$, so $R = f_p(0)^3 = 1$. Then $p \equiv 7 \pmod{9}$.
- $m \equiv 1 \pmod{3}$: $\psi = x(x^2 + 1)$. Let $\alpha^2 = -1$ in $\mathbb{F}_9$. Then $g(\alpha) = \alpha(\alpha^2 - 1) = \alpha$, so

$$R = (\alpha^{m+1} + 1)((-\alpha)^{m+1} + 1).$$

Odd $p = 3m + 1$ forces m even, hence $m + 1$ odd, and the product is $1 - (\alpha^2)^{m+1} = 1 - (-1) = 2$. Then $p \equiv 4 \pmod{9}$.

$r = 2$. Here $B_2 = x(x - 1)$ and $B'_2 = 2x - 1 = 2(x + 1)$ in $\mathbb{F}_3$, so

$$\psi = x(x - 1)(2(x + 1)^2 - m).$$

- $m \equiv 0 \pmod{3}$: $\psi = 2x(x - 1)(x + 1)^2$, all roots in $\mathbb{F}_3$, so $f_p(\beta) = 1$ at every root and $R = 2^p \equiv 2$. Then $p \equiv 2 \pmod{9}$.
- $m \equiv 2 \pmod{3}$: $\psi = 2x^2(x - 1)^2$, likewise $f_p(\beta) = 1$ at every root and $R = 2^p \equiv 2$. Then $p \equiv 8 \pmod{9}$.
- $m \equiv 1 \pmod{3}$: $\psi = 2x(x - 1)(x^2 + 2x + 2)$. The quadratic is irreducible ($\text{disc} \equiv 2$, not a square). For a root β one has $\beta^2 = \beta + 1$, $g(\beta) = \beta(\beta^2 - 1) = \beta^2$, and $B_2(\beta) = \beta^2 - \beta = 1$, so $R = 2^p N_{\mathbb{F}_9/\mathbb{F}_3}(1 + (\beta^2)^m)$. Now $(\beta^2)^2 = (\beta + 1)^2 = 2$, hence β^2 has order 4 in $\mathbb{F}_9^\times$. Odd $p = 3m + 2$ forces m odd, so $(\beta^2)^m \in \{\beta^2, -\beta^2\}$; both $1 + \beta^2 = \beta - 1$ and $1 - \beta^2 = -\beta$ have norm 2. Thus $R = 2^p \cdot 2 \equiv 1$. Then $p \equiv 5 \pmod{9}$.

The Kronecker symbol is $((-1)^{(p-1)/2}R/3)$. Since $(1/3) = 1$ and $(2/3) = -1$, this is $(-1)^{(p-1)/2}$ when $R = 1$ and the opposite when $R = 2$. Combining the class modulo 9 with the sign modulo 4 yields the six listed classes modulo 36, and no others among the units of $\mathbb{Z}/36\mathbb{Z}$. $\square$

The good set is stable under $p \mapsto -p$ (this fails already for $q = 5$). Dirichlet's theorem supplies infinitely many primes in each of the six classes, and Corollary 10 with $q = 3$ yields $\text{Gal}(f_p/\mathbb{Q}) = S_p$ on a set of Dirichlet density $\frac{1}{2}$. In particular:

Theorem 15. *There are infinitely many primes p with $\text{Gal}(f_p/\mathbb{Q}) = S_p$.*

This theorem is computer-free: it uses only Theorem 3 with Lemma 4 (the two finite checks in that proof are hand-verifiable), Proposition 14, and Dirichlet's theorem.

Each further prime q contributes its own periodic good set, of some Dirichlet density ε_q . The first three values are exact:

$$\varepsilon_3 = \frac{1}{2}, \quad \varepsilon_5 = \frac{11}{20}, \quad \varepsilon_7 = \frac{323}{648}.$$

(The last is 18088 of the 36288 units modulo 134064.) For $q = 5$, an inspection of the twenty fibres of u_r shows that the part coprime to g has degree 0, 2, or 4. Lemma 11 with $d \leq 4$, using Remark 12 for $d = 2$, yields period dividing $15600 = \text{lcm}(4, 5^2, 5^4 - 1)$. Evaluating (1) at one representative of each odd class modulo 15600 shows that the period divides 600, and no proper even divisor of 600 works. Among the 160 units of $\mathbb{Z}/600\mathbb{Z}$, 88 have symbol -1 , hence $\varepsilon_5 = \frac{11}{20}$. Combining with Proposition 14: of the 480 units of $\mathbb{Z}/1800\mathbb{Z}$, $372 = 240 + 264 - 132$ are good for $q = 3$ or $q = 5$ — the six classes modulo 36 lift to 240 units modulo 1800, the 88 classes modulo 600 lift to 264, and the intersection has 132 units. As $132 = 480 \cdot \frac{1}{2} \cdot \frac{11}{20}$, the two good sets are exactly independent on the units modulo 1800, which is not automatic since $\gcd(36, 600) = 12$.

The pattern does not continue. Evaluating (1) at every one of the 725,760 odd unit classes modulo $\text{lcm}(36, 600, 134064) = 3,351,600$ gives the exact joint law of the three symbols. The marginals come out at $\frac{1}{2}, \frac{11}{20}, \frac{323}{648}$, reproducing the values above on a modulus 25 times larger; the pair $\{3, 5\}$ is again exactly independent; but $\{3, 7\}$ is not. One has

$$\Pr[s_3 = -1, s_7 = -1] = \frac{787}{3024} = 0.26025\dots, \quad \varepsilon_3\varepsilon_7 = \frac{323}{1296} = 0.24922\dots,$$

so the good sets at $q = 3$ and $q = 7$ are positively correlated and the product formula $1 - \prod(1 - \varepsilon_q) = \frac{511}{576}$ is wrong.

Theorem 16. *The set of primes p with $(\text{disc } f_p/3) = -1$ has Dirichlet density $\frac{1}{2}$; adjoining $q = 5$ gives $\frac{31}{40}$; adjoining $q = 7$ gives*

$$\frac{7117}{8064} = 0.88256\dots$$

On each of these sets, $\text{Gal}(f_p/\mathbb{Q}) = S_p$.

These are the exact joint coverage densities computed here. Further enumeration is limited by the growth of the bounds in Remark 13.

The least-witness statistics over the 664577 primes $5 \leq p < 10^7$ exhibit the joint law rather than merely the marginals. For $q \in \{3, 5, 7\}$ the least witness is a function of p modulo 3351600, and it is the same function that the class enumeration tabulates; so what follows is not a second evaluation of the symbols but a check that the primes sample that class function faithfully, and a demonstration that the correlation can be read off the witness list given only $\varepsilon_3, \varepsilon_5, \varepsilon_7$. Writing P_q for the proportion of primes whose least witness is q :

	$q = 3$	$q = 5$	$q = 7$
observed	0.50003	0.27483	0.10767
joint law	$\frac{1}{2} = 0.50000$	$\frac{11}{40} = 0.27500$	$\frac{4337}{40320} = 0.10756$
discrepancy	0.05σ	0.31σ	0.27σ

Had the three symbols been independent, P_7 would be $(1 - \varepsilon_3)(1 - \varepsilon_5)\varepsilon_7 = 0.11215$, which differs from the observed fraction by 11.8 binomial-model standard deviations. That the symbols are correlated is not news — Theorem 16 proves it — but the discrepancy shows the correlation is legible in the witness list itself, at a sample size where the two models are far apart; below 10^5 they differ by only about 1.5σ . These standard-deviation figures are descriptive diagnostics under a binomial model, not probability bounds for the deterministic set of primes or a proof of an asymptotic sampling law. Likewise $P_3 + P_5 + P_7 = 0.882525$ against the predicted $\frac{7117}{8064} = 0.882564$, a discrepancy of 0.1σ .

Remark 17 (Ramification is rare, and computable). The same computation exhibits a set of unit classes of density $\frac{1}{324}$ on which $s_7 \equiv 0$: every prime in them has $7 \mid \text{disc } f_p$. No such classes occur at $q = 3$ or $q = 5$. Ramification of a fixed ℓ is therefore periodic in p like the symbol itself, and — unlike the symbol — its density can be computed without knowing the period. Write δ_ℓ for the density, among odd integers n coprime to ℓ , of those with $\ell \mid \text{disc } f_n$; this is the quantity the fibres compute. This integer density must be distinguished from the density among primes, which is obtained by restricting to unit classes and renormalising. Such conditioning need not decrease a proportion in general. At $\ell = 7$, the prime density is $1/324$, whereas $\delta_7 = 50/3591$. For either population separately, the event $v_\ell = 1$ is contained in $v_\ell \geq 1$; only the former supplies the simple-ramification certificate.

A factor of the product in (1) vanishes exactly when $g(\beta)^m = -1/B_r(\beta)$, and only β of degree ≥ 2 can contribute, since $\beta \in \mathbb{F}_\ell$ gives $g(\beta) = 0$ and a factor 1, while $B_r(\beta) \neq 0$ because B_r splits over $\mathbb{F}_\ell$. The exponent is not free: it is pinned modulo ℓ by the fibre $u_r(\beta) = m$ and modulo 2 by p odd, so m runs over a single class M modulo 2ℓ and $g(\beta)^m$ over the coset $\gamma^M \langle \gamma^{2\ell} \rangle$, $\gamma = g(\beta)$. Writing $o = \text{ord } \gamma$ and $e = \gcd(o, 2\ell)$, the group $\langle \gamma \rangle$ is the unique subgroup of its order in the cyclic group $\mathbb{F}_{\ell^d}^\times$, so $\langle \gamma^{2\ell} \rangle = \langle \gamma^e \rangle$ and membership is the single test $x^{o/e} = 1$. The vanishing class then has conditional density e/o , whence

$$\delta_\ell \leq \frac{1}{\ell(\ell-1)} \sum \frac{e}{o},$$

a density over odd n coprime to ℓ , the sum running over those r , m_0 and irreducible factors of $u_r - m_0$ of degree ≥ 2 for which $(-B_r(\beta)^{-1} \gamma^{-M})^{o/e} = 1$. This is a finite computation for each ℓ , independent of the period — which matters, since by Remark 13 the period is out of reach for $\ell \geq 11$.

The inequality is the honest statement, because ramification is the event that *some* factor vanishes while the right-hand side adds one term per factor. Distinct fibres never overlap: two of them differ either in $r = p \bmod \ell$ or in $m \bmod \ell$. So the only possible overcount is between two factors of one fibre (r, m_0) , and equality holds as soon as each fibre contributes at most one passing factor. That is the case for $\ell = 7$ and $\ell = 11$, where the passing factors lie in 12 and 32 distinct fibres respectively; from $\ell = 13$ on some fibres carry two or three, and the displayed sum is then an upper bound. Its

difference from the union density depends on the intersections of vanishing classes within each fibre; no uniform asymptotic error estimate is used here.

At $\ell = 7$, where equality holds, the formula gives $\delta_7 = \frac{50}{3591}$, agreeing exactly with an enumeration over the full period 134064. At $\ell = 3$ and $\ell = 5$ the sum is empty, so *3 and 5 never divide disc f_p , for any p* ; for $\ell = 3$ that is already visible in Proposition 14, where $R \in \{1, 2\}$.

The finite sample $5 \leq p < 10^5$ provides a separate observation. Summing the measured fractions of ramified primes over odd $\ell \leq 199$, with each eligible population restricted to $p > \ell$, gives approximately 0.0072. These measured prime proportions are not the integer densities δ_ℓ , and their finite sum does not bound the union over all ramifying primes. In that sample no ramification is observed at $\ell = 11, 13, 19, 31, 43$, but this does not imply their exclusion for every p . The data and denominators are recorded in `ancillary/sweep_results.txt`. The small-discriminant computations in §8 find simple ramification at primes as large as 669577; they make no uniform assertion about all degrees.

7. A CFSG-FREE CERTIFICATE

Theorem 3 buys its conclusion with Guralnick's theorem. For f_p there is a second, entirely classical certificate, which uses no classification at all. It is the standard Jordan–Dedekind test of computational Galois theory [20, 17, 19]; what is new here is only that it fires for this family, and that recording the witness makes the identification independent of §5 and of CFSG.

Theorem 18. *Let $f \in \mathbb{Z}[x]$ be monic irreducible of prime degree $p \geq 7$, and let q be a prime such that $f \bmod q$ is separable, with irreducible factors of degrees $\ell_1, \dots, \ell_k$. Suppose*

- (1) *some prime $\ell \in [3, p-3]$ occurs exactly once among the ℓ_i and is coprime to every other ℓ_j , and*
- (2) *$p-k$ is odd.*

Then $\text{Gal}(f/\mathbb{Q}) = S_p$.

Lemma 19 (Isolation; [17, Comments 1]). *Let σ have cycle type $(\ell_1, \dots, \ell_k)$ and suppose $\ell = \ell_i$ is coprime to $N = \text{lcm}_{j \neq i} \ell_j$. Then σ^N is an ℓ -cycle.*

Proof. Raising to the power N kills every cycle of length ℓ_j , $j \neq i$. On the remaining cycle, $\gcd(\ell, N) = 1$, so σ^N restricts to a generator of that cycle's cyclic group; it is an ℓ -cycle with $p - \ell$ fixed points. $\square$

Proof of Theorem 18. Let $G = \text{Gal}(f/\mathbb{Q}) \leq S_p$, transitive because f is irreducible, hence primitive: a block system has blocks of size dividing p . Separability of $f \bmod q$ gives $q \nmid \text{disc } f$, so q does not divide the index $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$ and Dedekind's theorem [14, Thm. 1] applies: $\sigma = \text{Frob}_q$ has cycle type $(\ell_1, \dots, \ell_k)$. By Lemma 19, σ^N is an ℓ -cycle with ℓ prime and $3 \leq \ell \leq p-3$, so Jordan's theorem [21, Thm. 13.9] gives $A_p \leq G$. Finally $\text{sgn } \sigma = \prod_i (-1)^{\ell_i-1} = (-1)^{p-k} = -1$, so $G \not\leq A_p$ and $G = S_p$. $\square$

The two certificates are complementary and their hypotheses can overlap. For this family, Theorem 3 is applied to a σ with no fixed point, which

for f_p is automatic at every $q \leq p$ but says nothing about the rest of the cycle type; Theorem 18 needs an isolated cycle of prime length $\leq p - 3$, a condition on the cycle type that is unrelated to the fixed-point count. An odd derangement with an isolated prime cycle can satisfy both tests. Both require σ odd, and neither can be replaced by the other.

Remark 20 (Affine types never certify). A non-identity affine map is a p -cycle or has type $(1, d^{(p-1)/d})$ with $d = \text{ord}(a) > 1$. In the first case $\ell = p > p - 3$; in the second, a cycle of length d is isolated only if $(p-1)/d = 1$, i.e. $d = p - 1$, which is even and exceeds $p - 3$. So no affine element satisfies hypothesis (1), consistently with Jordan's theorem.

Remark 21 (Why this does not settle every p). Chebotarev describes the useful primes q after the Galois group is known, so an S_p cycle-type prediction cannot itself prove Conjecture 1. Unger's theorem [20, Thm. 2] shows that almost all permutations power to a prime-length cycle. The present test additionally requires an isolated prime length in $[3, p - 3]$ and an odd permutation. Its unconditional density among all Frobenius classes is at most $1/2$, since exactly half of S_p is odd. We do not infer a density tending to one, or a uniform performance comparison with the smaller-prime discriminant test. Both the restrictions on cycle length and the range of eligible primes must be considered in such a comparison.

Hypothesis (1) is checked by a distinct-degree factorisation of $f_p \bmod q$, which returns the multiset $\{\ell_i\}$ without producing the factors. For every prime $7 \leq p < 1500$ a witness exists with $q \leq 61$; the largest least odd-prime witness is $q = 61$, attained at $p = 1301$, and $q = 3$ already succeeds for 80 of the 236 primes in the range, $q \leq 7$ for 169 of them. The witnesses are listed in `ancillary/jordan_witnesses.txt`, one row $(p, q, \ell, \{\ell_i\})$ per prime, and each row is re-derived from (p, q) alone by `tools/verify_jordan.py`. For those p , the identification $\text{Gal}(f_p/\mathbb{Q}) = S_p$ rests on Dedekind and Jordan alone.

8. CLOSED ROUTES

Proposition 22 (Totally real). *For every prime $p \geq 5$, f_p has p real roots. (For $p = 3$ there is one real root.)*

Proof. Write $x_k = k - \frac{1}{2}$ for $0 \leq k \leq p$. The sign of $\varphi(x_k)$ is $(-1)^{p-k}$, and

$$\frac{|\varphi(x_{k+1})|}{|\varphi(x_k)|} = \frac{k + \frac{1}{2}}{p - k - \frac{1}{2}}.$$

Writing $m = (p - 1)/2$, the sequence $|\varphi(x_k)|$ decreases toward its equal minima at $k = m, m + 1$ and then increases. Its minimum is

$$\min_k |\varphi(x_k)| = \frac{(2m - 1)!! (2m + 1)!!}{2^p}.$$

This equals $45/32 > 1$ at $p = 5$ and thereafter grows by the factor $(2m + 1)(2m + 3)/4$. Hence $f_p = \varphi + 1$ has the sign of φ at all $p + 1$ half-integers, so p sign changes and p real roots. At $p = 3$ the minimum is $3/8 < 1$. $\square$

Complex conjugation is therefore not a transposition for $p \geq 5$, and Conrad's criterion (irreducible of prime degree, exactly two non-real roots) does not apply.

Proposition 23. *The polynomial φ is a Morse function in the sense of Serre [11, §4.4]: the zeros of φ' are simple and the finite critical values are pairwise distinct. Consequently $\text{Gal}(\varphi - t/\mathbb{Q}(t)) = S_p$ [11, Thm. 4.4.5].*

Proof. On each interval $(k, k+1)$, $k = 0, \dots, p-2$, the logarithmic derivative $H(x) = \sum_{j=0}^{p-1} (x-j)^{-1}$ is strictly decreasing ($H' = -\sum (x-j)^{-2} < 0$), so there is a unique simple zero c_k . Thus $\varphi''(c_k) = \varphi(c_k)H'(c_k) \neq 0$. (The point ∞ is a totally ramified point of ramification index p , excluded from Serre's definition.) The involution $x \mapsto (p-1) - x$ sends c_k to c_{p-2-k} and φ to $-\varphi$.

Write $\mu_k = \max_{(k, k+1)} |\varphi|$ and $m = (p-1)/2$. For $x \in (k, k+1)$ and $k \leq m-2$, the ratio satisfies the uniform bound

$$\left| \frac{\varphi(x+1)}{\varphi(x)} \right| = \frac{x+1}{p-1-x} < \frac{k+2}{p-k-2} \leq \frac{m}{m+1} < 1.$$

Taking maxima therefore gives $\mu_{k+1} \leq \frac{m}{m+1} \mu_k < \mu_k$ for $k = 0, \dots, m-2$. The central pair has $\mu_{m-1} = \mu_m$ by the involution, so equal magnitudes occur only for mirror pairs $c_k \leftrightarrow c_{p-2-k}$. Those indices have opposite parity (p odd), hence opposite signs $(-1)^{p-1-k}$; since no critical value vanishes (the roots of φ are simple), the two members of a mirror pair are distinct. Critical values of different magnitude are distinct a fortiori, so all finite critical values are pairwise distinct. $\square$

Hilbert irreducibility gives many specialisations with group S_p , but does not decide the prescribed value $t = -1$.

For every integer a with $0 \leq a < p$, the polynomial $f_p(x+a)$ is monic with constant term 1. Its Newton polygon at any prime is horizontal, so these translates do not provide the nonzero slopes used by Newton-polygon methods such as [16]. This observation concerns those particular translates, not every possible change of variables or basis. Theorem 18 instead obtains cycles from factorization modulo unramified primes. Generic specialisation results do not determine the fixed value $t = -1$.

The arithmetic argument in [11, Remark 2, pp. 42–43] uses reductions with at most one double root and all other roots simple: the inertia groups are generated by transpositions, and Minkowski's theorem makes them generate Gal . Serre's worked example is $x^n - x - 1$, whose irreducibility he attributes to Selmer [8]. Related trinomial Galois-group criteria are given by Nart–Vila [9] and Osada [10]. For f_p itself, squarefreeness of $\text{disc } f_p$ is a sufficient condition for the same inertia argument. Direct factorisation gives this for $p \leq 7$; already at $p = 97$ the discriminant has 12642 digits. Among $p \leq 97$, every ramified prime $\ell \leq 10^6$ has $v_\ell(\text{disc } f_p) = 1$. A single such ℓ produces an inertia transposition and, with the mod- p p -cycle, yields S_p ; this fires for 18 of the 24 primes $p \leq 97$.

9. THE FAMILY $(x)_p + c$

The general certificates in §3 and §7 apply beyond the constant 1. Writing

$$f_{p,c}(x) = \varphi(x) + c = x(x-1) \cdots (x-p+1) + c, \quad c \in \mathbb{Z} \setminus \{0\},$$

the certificates apply verbatim, and the two-parameter family shows what is special about $c = 1$: it lies within a totally real regime. In fact every nonzero c with $|c| < \mu_{m-1}$ is totally real; $c = 1$ is not the only such constant.

Proposition 24. *Let p be an odd prime and $c \in \mathbb{Z}$ with $p \nmid c$.*

- (1) *$f_{p,c}$ is irreducible over $\mathbb{Q}$, and Frob_p is a p -cycle.*
- (2) *If $q \leq p$ is prime and $q \nmid c$, then $f_{p,c}$ has no root in $\mathbb{F}_q$.*

Consequently, if some odd prime $q < p$ with $q \nmid c$ satisfies $\left(\frac{\text{disc } f_{p,c}}{q}\right) = -1$, then $\text{Gal}(f_{p,c}/\mathbb{Q}) = S_p$.

Proof. (1) In $\mathbb{F}_p[x]$ one has $\varphi = \prod_{k \in \mathbb{F}_p} (x - k) = x^p - x$, so $f_{p,c} \equiv x^p - x + c \pmod{p}$. By Artin-Schreier, $x^p - x - a$ is irreducible over $\mathbb{F}_p$ exactly when $a \notin \varphi(\mathbb{F}_p) = \{0\}$; here $a = -c \neq 0$. The reduction is irreducible, hence separable, so $p \nmid \text{disc } f_{p,c}$ and Frob_p is a p -cycle. (2) The $p \geq q$ consecutive factors of $\varphi(a)$ run over every residue in $\mathbb{F}_q$, so $\varphi(a) = 0$ and $f_{p,c}(a) = c \neq 0$ in $\mathbb{F}_q$. The last statement is Theorem 3 applied with hypothesis (1) supplied by (2). $\square$

The symbol-periodicity calculus of §5 also carries over when $q < p$ and $q \nmid c$. Indeed $f'_{p,c} = g^{m-1} B_r(u_r - m)$ modulo q , and

$$\text{Res}(g, f_{p,c}) = c^q, \quad \text{Res}(B_r, f_{p,c}) = c^r.$$

Thus the analogue of (1) contains the prefactor $c^{q(m-1)+r} = c^{p-q}$, as well as replacing each $+1$ in the product by $+c$. Since $p-q$ is even, this nonzero prefactor is a square and disappears from the quadratic symbol. Consequently the same period bound works for the symbol at a fixed constant c , although the discriminant formula itself changes.

We now show that for $|c|$ in one explicit window the identification is elementary — no Frobenius, no classification, no computation.

Lemma 25 (Root count). *Let $p \geq 5$ be prime, let $c_0 < \cdots < c_{p-2}$ be the critical points of φ from Proposition 23, and put $\mu_k = |\varphi(c_k)|$. For every integer $c \neq 0$ whose absolute value is not a critical magnitude,*

$$\#\{x \in \mathbb{R} : f_{p,c}(x) = 0\} = 1 + 2 \#\{k \text{ odd} : |c| < \mu_k\}.$$

Proof. On $(k, k+1)$ exactly $k+1$ of the factors of φ are positive and $p-1-k$ are negative, so φ has sign $(-1)^{p-1-k} = (-1)^k$ there, p being odd; thus $v_k := \varphi(c_k)$ has sign $(-1)^k$, and consecutive critical values have opposite signs. By Proposition 23 the c_k are the only critical points, so φ is strictly monotone on each of the p intervals $(-\infty, c_0), (c_0, c_1), \dots, (c_{p-2}, \infty)$, with $\varphi(x) \rightarrow \pm\infty$ as $x \rightarrow \pm\infty$. A level L is attained exactly once on each such interval on which it lies strictly between the endpoint values, and not at all otherwise.

Take first $c > 0$, so $L = -c < 0$. The leftmost interval always contributes, since $v_0 > 0 > L$. For $1 \leq j \leq p-2$ the interval (c_{j-1}, c_j) contributes iff L lies strictly between v_{j-1} and v_j ; these have opposite signs, so the

span is (v_o, v_e) with o the odd and e the even index in $\{j-1, j\}$, and $L < 0 < v_e$ reduces the condition to $L > v_o$, i.e. $c < \mu_o$. The rightmost interval contributes iff $L > v_{p-2}$, i.e. $c < \mu_{p-2}$, and $p-2$ is odd. Now count. There are p monotonicity intervals: the leftmost, which always contributes; the rightmost; and the $p-2$ bounded ones. The leftmost is unconditional, and the remaining $p-1$ each carry the condition $|c| < \mu_k$ for one odd index k . Each odd k occurs for exactly two of them, namely (c_{k-1}, c_k) and (c_k, c_{k+1}) , the second of these being the rightmost interval when $k = p-2$. As there are $(p-1)/2$ odd indices in $\{0, \dots, p-2\}$, this accounts for $2 \cdot (p-1)/2 = p-1$ intervals, and $p = 1 + 2 \cdot \frac{p-1}{2}$ in total. The count follows.

For $c < 0$, the involution $x \mapsto (p-1) - x$ sends φ to $-\varphi$, hence $f_{p,c}((p-1) - x) = -f_{p,-c}(x)$; the two polynomials have the same number of real roots, so the count depends only on $|c|$. $\square$

Theorem 26 (Critical window). *Let $p \geq 5$ be prime, $m = (p-1)/2$, and let $c \in \mathbb{Z}$ satisfy $p \nmid c$ and*

$$\mu_{m-1} < |c| < \mu_{m-2}.$$

Then $f_{p,c}$ has exactly two non-real roots, and $\text{Gal}(f_{p,c}/\mathbb{Q}) = S_p$.

Proof. Proposition 23 gives $\mu_k = \mu_{p-2-k}$ and $\mu_0 > \mu_1 > \dots > \mu_{m-1} = \mu_m$. Since $p-2$ is odd, the involution $k \mapsto p-2-k$ reverses parity, so it matches each odd index with an even one; the odd indices therefore meet each mirror pair exactly once, and

$$\{\mu_k : k \text{ odd}\} = \{\mu_{m-1} < \mu_{m-2} < \dots < \mu_0\}$$

with each value occurring once. By Lemma 25, $f_{p,c}$ has $p-2j$ real roots, where $j = \#\{k \text{ odd} : \mu_k < |c|\}$; the stated window is precisely the condition $j = 1$. So $f_{p,c}$ has $p-2$ real roots and one pair of complex conjugate roots.

By Proposition 24, $f_{p,c}$ is irreducible of prime degree p , so $G = \text{Gal}(f_{p,c}/\mathbb{Q})$ is transitive and $p \mid |G|$; by Cauchy, G contains a p -cycle (or take Frob_p). Complex conjugation fixes the $p-2$ real roots and swaps the remaining pair, so G contains a transposition. Since p is prime, G is moreover primitive, and a primitive group containing a transposition is the full symmetric group; equivalently, a p -cycle together with a transposition generates S_p [1, Thm. 3.2]. Primality is used here as well: in composite degree a transitive group with a transposition need not be S_n [1, Rem. 3.3]. $\square$

Remark 27 (Why $c = 1$ is the hard case). The window is never reached by $c = \pm 1$. Indeed $\mu_k \geq |\varphi(k + \frac{1}{2})| \geq \min_j |\varphi(j - \frac{1}{2})| = (2m-1)!!(2m+1)!!/2^p$, which is $45/32 > 1$ at $p = 5$ and increases thereafter (Proposition 22); so $|c| = 1 < \mu_{m-1}$, $j = 0$, and f_p is totally real. That is Proposition 22 again, read off Lemma 25. The content of Theorem 26 is that the obstruction is quantitative and narrow: the conjugation route is closed for $c = 1$ only because 1 lies below every critical magnitude, and it reopens as soon as $|c|$ crosses the smallest of them. The whole apparatus of §§3–5 is the price of the single value $c = 1$.

Remark 28 (Only the first window). Larger $|c|$ does not give a single transposition by this argument. If j odd indices lie below $|c|$, complex conjugation is a product of j disjoint transpositions, of sign $(-1)^j$; Conrad's

criterion needs $j = 1$. At the top, $|c| > \mu_0$ leaves one real root and $(p-1)/2$ transpositions, an even permutation when $p \equiv 1 \pmod{4}$, which does not even exclude A_p . The first window is precisely where complex conjugation is a single transposition, so the stated two-non-real-root criterion applies. Other windows require additional arguments; this does not rule out other certificates for their groups.

The windows are wide. Writing N_p for the number of *positive* integers c in the window with $p \nmid c$ — each of which is matched by $-c$, since Lemma 25 depends only on $|c|$:

p	μ_{m-1}	μ_{m-2}	N_p
5	1.4187	3.6314	2
7	12.359	23.149	9
11	4804.9	7042.3	2034
13	171688	236529	59854
17	$5.3330 \cdot 10^8$	$6.7948 \cdot 10^8$	137 582 410
19	$4.3057 \cdot 10^{10}$	$5.3437 \cdot 10^{10}$	9 833 286 697

The measured ratio μ_{m-2}/μ_{m-1} decreases from 2.56 at $p = 5$ to 1.24 at $p = 19$. Growth of the integer count follows from a bound on the width, rather than from this table. For $x \in (m-2, m-1)$,

$$\left| \frac{\varphi(x+1)}{\varphi(x)} \right| = \frac{x+1}{2m-x} < \frac{m}{m+1}.$$

Taking maxima gives $\mu_{m-1} < m\mu_{m-2}/(m+1)$, so the width $W_p = \mu_{m-2} - \mu_{m-1}$ exceeds μ_{m-1}/m . Proposition 22 gives

$$A_m := \frac{(2m-1)!!(2m+1)!!}{2^{2m+1}} \leq \mu_{m-1}.$$

$$N_p \geq (1 - 1/p)W_p - 2 > (1 - 1/p)\frac{A_m}{m} - 2.$$

Here an open interval of width W_p contains at least $W_p - 1$ integers and at most $W_p/p + 1$ multiples of p . Since

$$A_m = \frac{p!}{2^p} \frac{\binom{2m}{m}}{4^m} \sim \frac{p!}{2^p \sqrt{\pi m}},$$

the lower bound is superexponential. The elementary upper bound $N_p \leq p^p + 1$ follows from $|\varphi(x)| \leq p^p$ on its bounded critical intervals; hence $\log N_p = p \log p + O(p)$ as p tends through primes. At $p = 5$ the constants $c = 2, 3$ lie in the window, giving S_5 directly. The displayed table and exact Sturm root counts for representative constants are checked by `tools/verify_window.py`.

10. COMPUTATIONAL CERTIFICATES AND REPRODUCIBILITY

The repository contains the TeX source, a working-revision PDF, certificate data, and standalone Python checks. The main witness file has columns (p, q) , with the exceptional row $(5, 19)$ interpreted by its factorization certificate. The Jordan file records $(p, q, \ell, \{\ell_i\})$. The period certificate for $q = 7$ records the proven upper modulus, the complete class count, a table

checksum, and counterexamples to smaller admissible periods. The measured density table records integer counts of good, eligible, and ramified primes; its certificate density includes symbol-zero rows, and its conditional unramified balance excludes them.

The aggregate commands are

```
python tools/verify.py -quick
python tools/verify.py -full.
```

Quick mode runs deterministic smoke and reference checks. Full mode additionally audits every main and Jordan certificate, checks least-witness minimality for the main list, verifies complete period and joint-class computations, and recomputes the stored density counts. Both report failures through a nonzero exit status. The required Python dependencies and separate PDF build command are documented in the README.

The full witness audit uses the reduced evaluation of (1). A separate checker compares it with the direct degree- p resultant on deterministic samples, and checks negative, positive, and ramified zero symbols. These two evaluations share the polynomial arithmetic in `fpcore.py`; a further modest-degree reference check uses SymPy's independent polynomial operations. Dataset checks verify complete prime coverage, uniqueness, bounds, and the special row rather than merely checking whichever rows happen to be present.

The computational dependencies are distinct from the mathematical ones. The odd-Frobenius certificate uses the classification-based Lemma 4; the Jordan certificate uses classical prime-cycle group theory; the critical-window theorem uses irreducibility, a real-root count, and the primitive-group transposition criterion. The Lean supplement covers only the affine fixed-point lemma and is not a formalisation of these arithmetic or classification inputs. Computational verification supplies finite certificates for the stated ranges, not a proof of Conjecture 1.

11. WHAT REMAINS

Write $\text{disc } f_p = (-1)^{(p-1)/2} \text{Res}(\varphi', \varphi + 1)$. Reducing modulo p (where $\varphi' \equiv -1$) gives the congruence

$$\text{disc } f_p \equiv (-1)^{(p+1)/2} \pmod{p},$$

always a quadratic residue, as forced by the even p -cycle Frob_p . This is a consistency check, not a constraint on witnesses $q < p$.

For a fixed finite set Q of smaller-prime candidates, periodicity makes the certified primes a union of reduced residue classes. The joint law on a common period gives its density. Theorem 16 shows that the individual events cannot be multiplied as though they were independent. The data refute coverage by $q \leq 71$, because $p = 9683099$ has least witness 73. They do not prove that every fixed finite set fails, or that the least witness is unbounded.

A precise sufficient condition for density-one coverage can be stated on the finite uniform space of unit classes modulo a common period. Write $\mathcal{E}_q$

for the negative-symbol event and $X_Q = \sum_{q \in Q} \mathbf{1}_{\mathcal{E}_q}$. Then

$$\begin{aligned} \Pr(X_Q = 0) &\leq \frac{\text{Var}(X_Q)}{(\mathbb{E}X_Q)^2}, \\ \text{Var}(X_Q) &= \sum_{q \in Q} \varepsilon_q(1 - \varepsilon_q) \\ &\quad + 2 \sum_{q < q'} (\Pr(\mathcal{E}_q \cap \mathcal{E}_{q'}) - \varepsilon_q \varepsilon_{q'}). \end{aligned}$$

Consequently $\mathbb{E}X_Q \rightarrow \infty$, together with a covariance sum $o((\mathbb{E}X_Q)^2)$, would make the uncovered densities tend to zero. A divergent sum of marginal densities alone is insufficient: identical events give an immediate counterexample. No such covariance estimate is proved here. Even density-one coverage would leave open individual exceptions to Conjecture 1; for a specified p , the certificate requires $q < p$.

The fibre structure can help organise the marginal problem. Each q has $q(q-1)$ fibres (r, m_0) , the reduced classes modulo q^2 . The symbol in (1) is $\chi_q((-1)^{(p-1)/2}R)$. If R is a nonzero constant on a fibre, the sign factor can still vary when $q \equiv 3 \pmod{4}$; a zero residual gives the constant zero symbol. The exact small-prime computations distinguish a constant residue from its constant quadratic character; the latter can hold while the former varies. They give:

- $q = 3$: all 6 fibres have constant R , none have constant symbol, and $\varepsilon_3 = \frac{1}{2}$.
- $q = 5$: of 20 fibres, 8 have constant R , while 10 have constant $\chi_5(R)$; these 10 have constant symbol. Six are always -1 and four always $+1$; the remaining ten are balanced. Hence $\varepsilon_5 = (6 + 5)/20 = \frac{11}{20}$. Six of the ten constants are the fibres on which $u_r - m_0$ splits over $\mathbb{F}_5$, and the symbol is then $\chi_5(r)$: $(r, m_0) = (1, 0), (1, 4), (4, 0), (4, 4)$ with $\chi_5(r) = +1$, and $(2, 0), (3, 4)$ with $\chi_5(r) = -1$. The excess of two always- -1 fibres in twenty is the $+\frac{1}{20}$.
- $q = 7$: of 42 fibres, 6 have constant R , while 14 have constant $\chi_7(R)$; none have constant symbol. Among unramified unit classes one has exactly $\frac{1}{2}$; among all unit classes, $\varepsilon_7 = \frac{323}{648}$, the deficit $\frac{1}{648}$ being ramification (Remark 17).

For quadratic fibres, Remark 12 bounds the order of $g(\beta)$ by $2(q-1)$, so estimates over an entire extension field do not directly control these short cyclic orbits. The complete unit-class census is reproduced by

`python tools/verify_fibre_census.py.`

At the three small primes above the nonconstant unramified fibres are balanced. No trichotomy $\{0, 1/2, 1\}$ is asserted for all q , and mere nonconstancy does not give a uniform lower bound on the negative fraction. An argument for the marginal densities would need a quantitative estimate of their weighted mean; an argument for limiting coverage would also need information about their joint law.

The $q = 3$ calculation reduces any search for a smaller-prime nonresidue to the complementary prime classes $1, 5, 11, 25, 31, 35 \pmod{36}$; adding $q = 5, 7$ further reduces their density. The resulting tests are sufficient conditions,

not an equivalence with Conjecture 1. They certify all 664576 primes $7 \leq p < 10^7$; the extra $p = 5$ row and separate $p = 3$ argument complete the finite odd-prime identification.

Acknowledgement of assistance. AI assistants were used in drafting and computational development, including this revision. Grok also reviewed the classification argument, critical-window proof, reduced-discriminant identity, and bibliography. The author is Open; the assistance does not constitute an independent referee report or a formal proof of the entire paper.

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